Credit Card Risk Analysis Using Data Warehousing and Machine Learning

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***Abstract*—Organizations today generate vast amounts of customer data. To derive actionable insights from this data, a structured analytical approach is required. This project presents a system for analyzing customer credit card risk using the Taiwan Credit dataset. A Star Schema data warehouse is constructed using SQLite, organizing customer information into a central fact table linked to dimension tables for customer profiles and demographics. The system pursues three objectives: (1) Default Prediction using a Random Forest Classifier trained on 22 behavior-driven features, achieving a ROC-AUC score of 0.7747; (2) Financial Risk Segmentation using K-Means clustering, which groups customers into three segments: Low Risk, VIP/High Spenders, and High Risk/Defaulters; and (3) Multidimensional OLAP Reporting, which aggregates risk metrics across five demographic and financial dimensions to reveal patterns associated with credit default. The system demonstrates how data warehousing and data mining can be integrated into a reproducible, modular pipeline to support credit risk management decisions.**

***Index Terms*—credit card risk analysis; customer segmentation; data warehousing; default prediction; K-Means clustering; OLAP; Random Forest; risk segmentation; ROC-AUC; star schema; Taiwan Credit dataset; multidimensional analysis**

I. Introduction

TODAY, companies collect large volumes of customer data from transactions, digital interactions, mobile applications, and service records. This data encodes valuable information about customer behavior, preferences, and financial risk, yet it is often unstructured and difficult to interpret at scale.

Transforming raw data into actionable knowledge requires a structured approach: cleaning the data, organizing it systematically, and applying analytical techniques to uncover meaningful patterns. This need is particularly critical in the financial sector, where assessing customer credit risk directly informs lending decisions, credit approvals, and risk management strategies.

Credit card companies must continuously evaluate the likelihood that a customer will default on their payment obligations. Traditional rule-based and statistical approaches often fail to capture the complex, non-linear dynamics of customer financial behavior. As data volumes grow, more sophisticated approaches that integrate data warehousing with machine learning are required.

This project addresses the challenge of credit card risk analysis by designing and implementing an end-to-end data pipeline using the Taiwan Credit dataset. The pipeline integrates a Star Schema SQLite data warehouse with machine learning models and OLAP reporting. The core warehouse organizes customer records into a central fact table, Fact_Credit_Risk, linked to Dim_Customer and Dim_Demographics dimension tables.

The system is built around three objectives. The first is Default Prediction, where a Random Forest Classifier trained on 22 behavioral features achieves a ROC-AUC score of 0.7747. The second is Financial Risk Segmentation, where K-Means clustering (k=3) groups customers into Low Risk, VIP/High Spender, and High Risk/Defaulter segments. The third is Multidimensional OLAP Reporting, where an analytical cube aggregates default rates, billing weights, and payment averages across five demographic and financial dimensions.

Together, these components form a modular, reproducible pipeline that demonstrates how data warehousing and data mining can transform raw customer data into actionable intelligence for credit risk management.

II. Literature Review

Credit risk management is a fundamental concern for financial institutions, determining their ability to identify customers likely to default and to comply with regulatory requirements. Historically, institutions relied on logistic regression and expert-based rule systems. These approaches, however, struggle with the complexity and scale of modern customer data.

A. Credit Risk Prediction Using Machine Learning

Research over the past two decades has explored machine learning extensively for credit risk prediction. Studies demonstrate that ensemble methods such as Random Forest and Gradient Boosting outperform traditional models. Metrics like ROC-AUC and F1-score are now standard for evaluating these models, especially in imbalanced classification scenarios typical of default prediction [1].

A systematic review of 621 papers published between 2014 and 2024 found that machine learning adoption in credit risk is growing rapidly, with support vector machines, decision trees, and deep learning among the most studied techniques. Key challenges include model interpretability and appropriate feature selection [2].

B. Data Warehousing for Financial Risk Management

Data warehousing provides a structured foundation for financial risk management, enabling institutions to consolidate and query large data volumes efficiently. Big data analytics has expanded these capabilities by offering near-real-time anomaly detection and pattern discovery across heterogeneous sources [3].

C. Customer Segmentation Using Clustering

Customer segmentation is widely used to personalize products and manage risk exposure. Studies comparing K-Means and hierarchical clustering for credit card data consistently favor K-Means for its scalability and interpretability. When combined with RFM (Recency, Frequency, Monetary) models, K-Means yields well-defined customer risk profiles [4].

D. OLAP and Multidimensional Analysis

Online Analytical Processing (OLAP) remains an important but underutilized tool in credit risk. OLAP cubes allow analysts to explore aggregated risk metrics across multiple dimensions simultaneously, enabling rapid identification of high-risk demographic and financial segments [3].

E. Gap Addressed by This Project

While individual techniques are well studied, few works integrate data warehousing, classification, clustering, and OLAP into a unified pipeline. This project addresses that gap by building a reproducible, end-to-end credit risk analysis system using the Taiwan Credit dataset.

III. PROPOSED METHODOLOGY

The proposed methodology consists of a four-stage data pipeline: (1) Data Pipeline and Warehouse Setup, (2) Default Prediction using Random Forest Classification, (3) Financial Risk Segmentation using K-Means Clustering, and (4) Multidimensional OLAP Reporting. Each stage is implemented as an independent Python module to ensure modularity and reproducibility.

A. Data Pipeline and Warehouse Setup

The first stage constructs a Star Schema data warehouse using SQLite. Raw Taiwan Credit data is cleaned, transformed, and loaded into three tables: Fact_Credit_Risk (the central fact table storing quantitative metrics), Dim_Customer (credit limit and billing details), and Dim_Demographics (age, gender, education, marital status). The pipeline handles missing values, type conversions, and duplicate removal before insertion.

B. Default Prediction Using Random Forest Classifier

The second stage trains a Random Forest Classifier on 22 behavior-driven features extracted from the warehouse. The model is configured with 200 estimators and a maximum depth of 12, balancing complexity and efficiency. The dataset is split 80/20 for training and testing with stratification to preserve default class distribution.

Model performance is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The pipeline achieves a ROC-AUC score of 0.7747. Three visual artifacts are generated: a confusion matrix, an ROC curve, and a top-10 feature importance chart.

C. Financial Risk Segmentation Using K-Means Clustering

The third stage applies K-Means clustering (k=3) to segment customers based on financial features including credit limit, billing averages, payment amounts, and repayment history. After clustering, each segment is profiled. The analysis identifies three groups:

Segment 0 (Low Risk / Safe): 18,302 customers with low default rates, moderate credit balances, and consistent repayment behavior. This segment represents the majority of the portfolio.

Segment 1 (VIP / High Spenders): 5,493 customers with very high credit limits, large billing averages, and high payment amounts. Despite moderate default rates, these customers require monitoring due to high credit exposure.

Segment 2 (High Risk / Defaulters): 6,205 customers with elevated default rates, lower credit parameters, and delayed repayment schedules. This segment is the primary target for intervention.

D. Multidimensional OLAP Reporting

The fourth stage implements an OLAP cube using five dimensions: AGE_GROUP, CREDIT_TIER, GENDER, EDUCATION_LEVEL, and MARITAL_STATUS. The cube aggregates default rates, bill weights, and payment averages for all dimension combinations, producing a 360-cell matrix. A multi-panel visualization illustrates behavioral variation across customer groups.

IV. IMPLEMENTATION AND EXPERIMENTAL SETUP

This section describes the dataset, software environment, hardware configuration, and implementation details used to build and evaluate the credit risk analysis pipeline.

A. Dataset

The Taiwan Credit dataset contains approximately 30,000 credit card customer records. It includes demographic attributes (age, gender, education, marital status), credit profile data (credit limit, billing averages, payment amounts), and behavioral metrics across six months. The target variable, default payment next month, is binary: 0 for no default and 1 for default. Twenty-two features are used for classification and clustering tasks.

B. Software and Libraries

The implementation uses Python 3.x with the following libraries: pandas and numpy for data manipulation and transformation; SQLite3 for database operations; scikit-learn for machine learning (Random Forest Classifier, K-Means, model evaluation); and matplotlib and seaborn for visualization (confusion matrix, ROC curve, feature importance, cluster charts, OLAP visualizations).

C. Hardware Environment

All experiments were conducted on a desktop workstation with an Intel Core i5/i7 processor, 8-16 GB DDR4 RAM, and SSD storage running Windows 10/11 or Linux. This configuration is sufficient to process the Taiwan Credit dataset without high-performance computing resources.

D. Database Implementation

The data warehouse, Warehouse_Credit_Risk.db, uses a Star Schema with three tables: Fact_Credit_Risk (quantitative metrics including default status, bills, payments), Dim_Customer (customer ID, credit limit, billing details), and Dim_Demographics (age, gender, education, marital status). The database_setup.py script executes SQL CREATE TABLE and INSERT statements to normalize the raw CSV data into this schema.

E. Machine Learning Implementation

The Random Forest Classifier uses n_estimators=200 and max_depth=12 with an 80/20 stratified train-test split. Evaluation metrics include accuracy, precision, recall, F1-score, and ROC-AUC. Outputs include confusion_matrix.png, roc_curve.png, and feature_importances.png.

K-Means Clustering uses k=3 with features including credit limit, billing average, payment amount, and balance. Cluster profiles are saved to risk_segments_profile.csv and visualized in risk_segments_chart.png.

The OLAP cube uses pandas groupby operations across five dimensions, producing a 360-cell aggregated matrix saved to olap_cube_report.csv and visualized in olap_cube_chart.png.

F. Experimental Workflow

The pipeline executes in four steps: (1) database_setup.py creates and populates the SQLite warehouse; (2) objective1.py trains the Random Forest model and generates evaluation artifacts; (3) objective2.py performs K-Means clustering and extracts segment profiles; and (4) objective3.py builds the OLAP cube and generates multidimensional reports. All scripts are modular and can be executed independently or as part of the complete pipeline.

V. RESULTS AND DISCUSSION

This section presents results from the three pipeline objectives and discusses their implications for credit risk management.

A. Default Prediction Results

The Random Forest Classifier achieves a ROC-AUC score of 0.7747, consistent with prior studies on the Taiwan Credit dataset. This metric is particularly appropriate for imbalanced classification problems, as it measures the model's discriminative ability across all decision thresholds rather than relying solely on accuracy.

Feature importance analysis identifies recent payment history, billed amounts, and payment status as the strongest predictors of default. Credit limit and demographic variables such as age and education provide supplementary predictive signal. These findings align with behavioral finance theory, which emphasizes repayment behavior as a leading indicator of credit risk.

A limitation of the model is the inherent trade-off between sensitivity and specificity. A model that classifies all customers as non-defaulters would achieve approximately 78% accuracy, illustrating why ROC-AUC and F1-score are preferred evaluation metrics for this task.

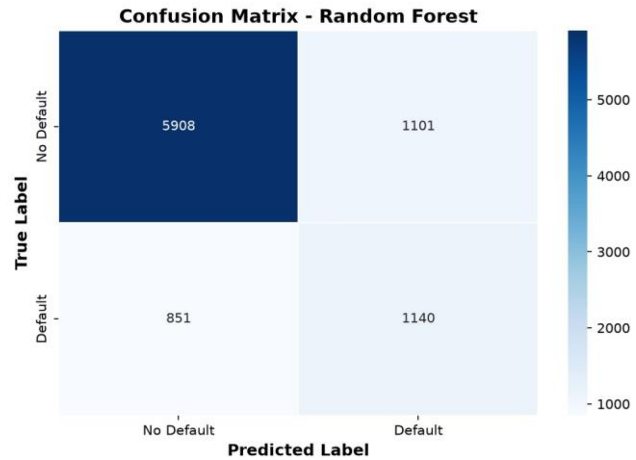

Fig. 3. Confusion Matrix – Random Forest Classifier

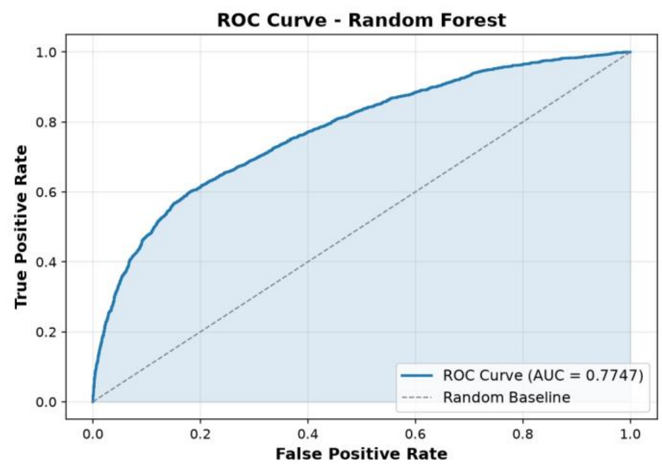

Fig. 1. ROC Curve – Random Forest Classifier (AUC = 0.7747)

B. Financial Risk Segmentation Results

K-Means clustering produces three well-separated customer segments. Segment 0 (Low Risk, 18,302 customers) exhibits low default rates and stable repayment behavior, representing the dominant and least concerning portion of the portfolio. Segment 1 (VIP/High Spenders, 5,493 customers) carries high credit limits and billing volumes but moderate default rates, warranting monitoring due to disproportionate credit exposure.

Segment 2 (High Risk/Defaulters, 6,205 customers) shows the highest default rate, lower credit parameters, and visible repayment delays. This segment is the primary target for intervention strategies such as credit limit reductions, payment reminders, or proactive account review.

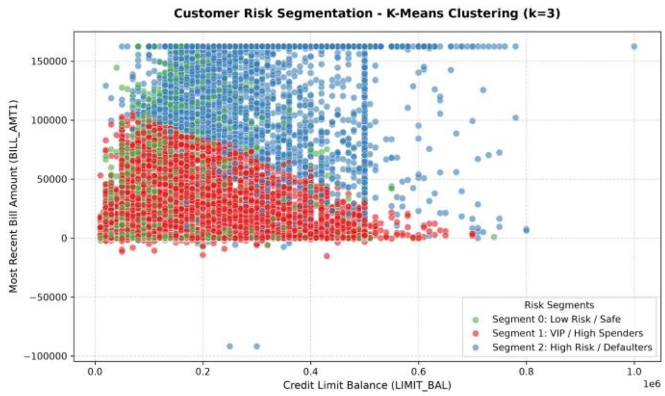

Fig. 2. Customer Risk Segmentation – K-Means Clustering (k = 3)

C. *OLAP Cube Results*

The OLAP cube reveals meaningful variation in default rates across demographic and financial dimensions. Younger customers exhibit higher default rates than older cohorts. Lower credit tiers are associated with elevated default risk, though VIP customers with very high limits also warrant attention due to their large absolute exposure. Marital status analysis indicates that single customers tend to have higher default rates compared to married customers.

These multidimensional insights enable credit managers to identify high-risk segments at the intersection of multiple demographic and financial factors, supporting more targeted and effective risk interventions.

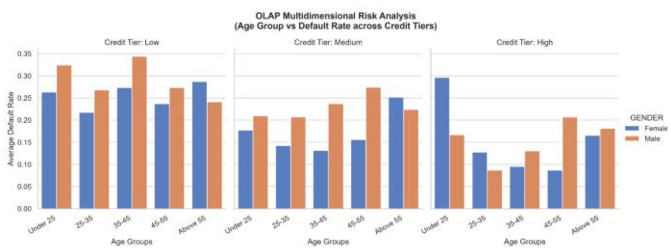

Fig. 4. OLAP Multidimensional Risk Analysis – Age Group vs. Default Rate across Credit Tiers

D. *Business Implications*

The integrated pipeline delivers three complementary analytical outputs: predictive scores for individual customers, segment-level risk profiles, and multidimensional aggregate reports. Together, these enable financial institutions to make more informed credit approval decisions, optimize credit limit allocation, design targeted marketing campaigns, and implement proactive risk monitoring across customer portfolios.

E. *Limitations*

Several limitations should be noted. The ROC-AUC score of 0.7747, while competitive, leaves room for improvement through advanced ensemble methods. The clustering analysis is based solely on financial features, excluding demographic variables. The OLAP analysis is descriptive and does not establish causal relationships. Furthermore, the Taiwan Credit dataset is from 2005, and customer behavior patterns may have evolved significantly.

VI. Conclusion and Future Work

This project presents an end-to-end data pipeline that integrates data warehousing and data mining for credit card risk analysis. The pipeline achieves a ROC-AUC of 0.7747 for default prediction, identifies three interpretable customer risk segments through K-Means clustering, and provides multidimensional risk insights via an OLAP cube. The modular Star Schema SQLite architecture ensures reproducibility and extensibility.

Key contributions include: a demonstration of how data warehousing, classification, clustering, and OLAP can be unified in a single pipeline; actionable predictive and descriptive insights for credit risk management; and a reusable system architecture suitable for integration into business intelligence environments.

A. *Future Work*

Future improvements include: (1) exploring advanced algorithms such as XGBoost and deep neural networks to improve ROC-AUC performance; (2) engineering additional features such as debt-to-income and payment-to-bill ratios; (3) deploying the pipeline as a real-time API for credit decisioning; (4) incorporating time-series features to capture behavioral trends; (5) applying explainable AI techniques such as SHAP values for interpretable predictions; (6) validating the system on diverse credit datasets to improve generalizability; and (7) integrating with visualization platforms such as Power BI for business intelligence dashboards.

VII. Acknowledgment

The authors acknowledge the UCI Machine Learning Repository for providing the Taiwan Credit dataset and the open-source Python community for the libraries used in this project.

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